

SeungWon Seo

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Professional Summary: Ph.D. Student at Korea University focusing on cooperative planning for embodied agents using large language models. Author of papers at CVPR, ICLR, AAAI (Oral), and SIGGRAPH, with research spanning uncertainty-aware reasoning, multi-agent systems, and VR-based character animation.

Education

Korea University Integrated MS-PhD, Computer Science and Engineering Seoul, Korea | Mar 2025 – Present
• Advisor: Prof. HyeongYeop Kang (IIIXR Lab).
Kyung Hee University B.E., Software Convergence, GPA: 3.82/4.5 Yongin, Korea | Mar 2020 – Feb 2024

Experience

NCSOFT Research Intern Seongnam, Korea | Jul 2023 – Aug 2023
• Researched text-based gesture animation generation.
IIIXR Lab Undergraduate Researcher Yongin, Korea | Dec 2021 – Feb 2024
• Conducted research on user interactions in VR environments.
Kyung Hee University Department Student Council President Yongin, Korea | Mar 2021 – Dec 2021
• Led departmental student council activities and organized various academic and social events.

Publications

**Equal contribution. †Corresponding author.*

Edit-As-Act: Goal-Regressive Planning for Open-Vocabulary 3D Indoor Scene Editing.

- SeongRae Noh, **SeungWon Seo**, Gyeong-Moon Park[†], and HyeongYeop Kang[†]
- CVPR 2026 | Denver, United States.

From Assumptions to Actions: Turning LLM Reasoning into Uncertainty-Aware Planning for Embodied Agents.

- SeungWon Seo**^{*}, SooBin Lim^{*}, SeongRae Noh^{*}, Haneul Kim, and HyeongYeop Kang[†]
- ICLR 2026 | Rio de Janeiro, Brazil.

REVECA: Adaptive Planning and Trajectory-based Validation in Cooperative Language Agents Using Information Relevance and Relative Proximity.

- SeungWon Seo**^{*}, SeongRae Noh^{*}, Junhyeok Lee, SooBin Lim, Won Hee Lee, and HyeongYeop Kang[†]
- AAAI 2025 | Philadelphia, USA. Oral Presentation

DAMO: A Deep Solver for Arbitrary Marker Configuration in Optical Motion Capture.

- KyeongMin Kim, **SeungWon Seo**, DongHeun Han, and HyeongYeop Kang[†]
- ACM Transactions on Graphics 2024 | Invited to SIGGRAPH 2025.

DARAM: Dynamic Avatar-Human Motion Remapping Technique for Realistic Virtual Stair Ascending Motions.

- SooBin Lim, **SeungWon Seo**, and HyeongYeop Kang[†]
- ACM SIGGRAPH 2023 | Los Angeles, USA

Teaching Assistant

Game Engine Basics [SWCON21200] Kyung Hee University | Sep 2024 – Dec 2024
Full-Stack Service Networking [SWCON49200] Kyung Hee University | Mar 2024 – Jun 2024

Honors and Awards

Master's Thesis Award – Honorable Mention Korea Computer Graphics Society | 2025
NC Fellowship – Top Prize NCSOFT | 2022
VR Game Development Competition – Top Prize Immersive Media Innovation Convergence University | 2022
UCPC Programming Contest – Finalist (28th place) Union of Clubs for Programming Contests | 2020

Technical Skills

Programming: Python, C++, C#
Machine Learning: PyTorch
LLM/Agent Systems: LangChain, LangGraph, vLLM
VR/Game: Unity 3D, Unreal Engine 5

Academic Service

Reviewer: ACM CHI 2025, ACM TOG 2025, SIGGRAPH Asia XR 2025
Student Volunteer: AAAI 2025

Languages

Korean (Native), English (Intermediate)